

Highlights

Manuscript Title: Positive-Class Weighting Induces Systematic Probability Distortion in Deep Learning Models for Chest Radiography under Deidentified-Data Cohort Shift

Target Journal: *Computer Methods and Programs in Biomedicine* (Elsevier)

- Positive loss weighting induces an additive population logit offset of $\log w$.
- Finite neural network logits empirically cluster tightly around theoretical $\log w$.
- Temperature scaling cannot correct this intercept-type calibration bias.
- Analytic offset ($z - \log w$) cuts 73.7% of Brier error without retraining.
- Selective triage exposes trade-offs between calibrated risk and screening referral.